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Linear Algebra ★

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Problem

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The NumPy module also comes with a number of built-in routines for linear algebra calculations. These can be found in the sub-module linalg.

linalg.det

The linalg.det tool computes the determinant of an array.

```
print numpy.linalg.det([[1 , 2], [2, 1]])    #Output : -3.0
```

linalg.eig

The linalg.eig computes the eigenvalues and right eigenvectors of a square array.

```
vals, vecs = numpy.linalg.eig([[1 , 2], [2, 1]])
print vals          #Output : [ 3. -1.]
print vecs          #Output : [[ 0.70710678 -0.70710678]
                        #      [ 0.70710678  0.70710678]]
```

linalg.inv

The linalg.inv tool computes the (multiplicative) inverse of a matrix.

```
print numpy.linalg.inv([[1 , 2], [2, 1]])    #Output : [[-0.33333333  0.66666667]
                        #      [ 0.66666667 -0.33333333]]
```

Other routines can be found [here](#)

Task

You are given a square matrix A with dimensions $N \times N$. Your task is to find the determinant. Note: Round the answer to 2 places after the decimal.

Input Format

The first line contains the integer N .

The next N lines contains the N space separated elements of array A .

Output Format

Print the determinant of A .

Sample Input

```
2
1.1 1.1
1.1 1.1
```

Sample Output

Author [\[deleted\]](#)Difficulty [Easy](#)

Max Score 20

Submitted By [102252](#)

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0.0

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Language

Pypy 3



1
2
3
4
5
6
7

Line: 7 Col: 1

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Congratulations

You solved this challenge. Would you like to challenge your friends?

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